Customer Behaviour & Personalisation in Fashion Retail

Complete Dissertation Data Analysis

This notebook analyses the primary survey data collected for the dissertation **Customer Behaviour & Personalisation in Fashion Retail**.

The workflow follows the methodology structure of the supplied reference notebook: data audit, cleaning, descriptive analysis, correlation, simple linear regression, regression assumptions, multiple linear regression, model comparison and business interpretation.

Important: The survey is cross-sectional. Therefore, the results identify statistical associations and should not be presented as proof of causation.

1. Methodology used in this notebook

The analysis includes:

1. Data loading and initial audit
2. Consent-based data cleaning
3. Missing-value assessment
4. Duplicate checking
5. Conversion of ordinal income, spending and purchase-frequency bands
6. Likert-scale coding from 1 to 5
7. Personalisation composite score
8. Brand-reputation composite score
9. Descriptive statistics
10. Interactive Plotly Express visualisations
11. Animated visualisations by age group
12. Pearson correlation
13. Correlation heatmap
14. Simple linear regression
15. Linearity, homoscedasticity and residual diagnostics
16. Shapiro-Wilk normality test
17. Durbin-Watson test
18. Multiple linear regression
19. VIF multicollinearity test
20. R^2, adjusted R^2 and AIC model comparison
21. Hypothesis testing
22. Dissertation-ready business interpretation

In [95]:
```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import statsmodels.api as sm
from scipy import stats
from statsmodels.formula.api import ols
from statsmodels.stats.stattools import durbin_watson
from statsmodels.stats.outliers_influence import variance_inflation
from pathlib import Path
from IPython.display import display

pd.set_option('display.max_columns', None)
print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

2. Load the dataset

The notebook searches for the Excel survey file automatically, avoiding machine-specific file paths.

In [96]:
```
df = pd.read_csv("Customer Behaviour & Personalisation in Fashion R
# Display first 5 rows
df.head()
```

Out[96]:

	Timestamp	Do you consent to participate in this research study?	What is your age?	What is your gender?	What is your monthly disposable income?	Approximately how much do you spend on fashion products each month?	How many fashion purchases do you typically make in a month?	Whi fashi produ catego do y purcha mo frequentl
0	21/07/2026 20:09:20	I agree to participate in this research study.	18–24	Male	£1,000–£1,999	£50–£99	1–2	Accessori
1	21/07/2026 20:11:03	I agree to participate in this research study.	18–24	Female	£2,000–£2,999	£200–£299	3–4	Clothi
2	22/07/2026 19:24:23	I agree to participate in this research study.	35–44	Female	Less than £1,000	Less than £50	3–4	Clothi
3	22/07/2026 19:25:35	I agree to participate in this research study.	18–24	Male	Less than £1,000	£50–£99	1–2	Clothi
4	22/07/2026 19:29:50	I agree to participate in this research study.	25–34	Male	£2,000–£2,999	£50–£99	3–4	Clothir Footwe

3. Initial data audit

This section checks the size, variables, data types and duplicate observations before analysis.

In [97]:
```
print("Dataset shape:", df.shape)
print("\nColumn names:")
for i, col in enumerate(df.columns, 1):
    print(f"{i}. {col}")

print("\nData types:")
```

```python
display(df.dtypes.to_frame("Data Type"))

print("\nDuplicate rows:", df.duplicated().sum())
```

```
Dataset shape: (70, 26)

Column names:
1. Timestamp
2. Do you consent to participate in this research study?
3. What is your age?
4. What is your gender?
5. What is your monthly disposable income?
6. Approximately how much do you spend on fashion products each mo
nth?
7. How many fashion purchases do you typically make in a month?
8. Which fashion product category do you purchase most frequently?
9. Which shopping channel do you use most frequently?
10. Which fashion brand do you purchase from most frequently?
11. What is the single most important factor influencing your fash
ion purchases?
12. To what extent do discounts influence your fashion purchasing
decisions?
13. To what extent do social media platforms or influencers influe
nce your fashion purchasing decisions?
14. To what extent do you agree with the following statements? [I
regularly receive personalised recommendations from fashion retail
ers.]
15. To what extent do you agree with the following statements? [Pe
rsonalised recommendations improve my shopping experience.]
16. To what extent do you agree with the following statements? [I
am more likely to purchase from a retailer that offers personalise
d recommendations.]
17. To what extent do you agree with the following statements? [I
feel valued when a retailer understands my shopping preferences.]
18. To what extent do you agree with the following statements? [Pe
rsonalised communication increases my loyalty to a fashion retaile
r.]
19. Please indicate your level of agreement with the following sta
tements. [I trust reputable fashion brands.]
20. Please indicate your level of agreement with the following sta
tements. [A brand's reputation influences my purchase decisions.]
21. Please indicate your level of agreement with the following sta
tements. [I would recommend a fashion brand with a good reputatio
n.]
22. Please indicate your level of agreement with the following sta
tements. [I am loyal to brands that meet my expectations.]
23. Please indicate your level of agreement with the following sta
tements. [A positive shopping experience improves my perception of
a brand.]
24. In your opinion, what could fashion retailers do to improve yo
ur shopping experience through personalisation?
25. Is there anything else you would like to share about your fash
ion shopping behaviour?
26. Email address

Data types:
```

	Data Type
Timestamp	object
Do you consent to participate in this research study?	object
What is your age?	object
What is your gender?	object
What is your monthly disposable income?	object
Approximately how much do you spend on fashion products each month?	object
How many fashion purchases do you typically make in a month?	object
Which fashion product category do you purchase most frequently?	object
Which shopping channel do you use most frequently?	object
Which fashion brand do you purchase from most frequently?	object
What is the single most important factor influencing your fashion purchases?	object
To what extent do discounts influence your fashion purchasing decisions?	float64
To what extent do social media platforms or influencers influence your fashion purchasing decisions?	float64
To what extent do you agree with the following statements? [I regularly receive personalised recommendations from fashion retailers.]	object
To what extent do you agree with the following statements? [Personalised recommendations improve my shopping experience.]	object
To what extent do you agree with the following statements? [I am more likely to purchase from a retailer that offers personalised recommendations.]	object
To what extent do you agree with the following statements? [I feel valued when a retailer understands my shopping preferences.]	object
To what extent do you agree with the following statements? [Personalised communication increases my loyalty to a fashion retailer.]	object
Please indicate your level of agreement with the following statements. [I trust reputable fashion brands.]	object
Please indicate your level of agreement with the following statements. [A brand's reputation influences my purchase decisions.]	object
Please indicate your level of agreement with the following statements. [I would recommend a fashion brand with a good reputation.]	object
Please indicate your level of agreement with the following statements. [I am loyal to brands that meet my expectations.]	object
Please indicate your level of agreement with the following statements. [A positive shopping experience improves my perception of a brand.]	object
In your opinion, what could fashion retailers do to improve your shopping experience through personalisation?	object
Is there anything else you would like to share about your fashion shopping behaviour?	object
Email address	float64

```
Duplicate rows: 0
```

4. Missing values

The raw dataset contains one non-consenting blank response. Because participation consent is a research requirement, this response is excluded rather than imputed.

The two open-ended questions are optional qualitative questions, so missing answers are labelled **No response**.

In [98]:
```
missing_report = pd.DataFrame({
    'Missing Values': df.isna().sum(),
    'Percentage': (df.isna().sum() / len(df) * 100).round(2)
}).sort_values('Percentage', ascending=False)

display(missing_report[missing_report['Missing Values'] > 0])
```

	Missing Values	Percentage
Email address	70	100.00
Is there anything else you would like to share about your fashion shopping behaviour?	44	62.86
In your opinion, what could fashion retailers do to improve your shopping experience through personalisation?	38	54.29
To what extent do you agree with the following statements? [Personalised recommendations improve my shopping experience.]	1	1.43
Please indicate your level of agreement with the following statements. [A positive shopping experience improves my perception of a brand.]	1	1.43
Please indicate your level of agreement with the following statements. [I am loyal to brands that meet my expectations.]	1	1.43
Please indicate your level of agreement with the following statements. [I would recommend a fashion brand with a good reputation.]	1	1.43
Please indicate your level of agreement with the following statements. [A brand's reputation influences my purchase decisions.]	1	1.43
Please indicate your level of agreement with the following statements. [I trust reputable fashion brands.]	1	1.43
To what extent do you agree with the following statements? [Personalised communication increases my loyalty to a fashion retailer.]	1	1.43
To what extent do you agree with the following statements? [I feel valued when a retailer understands my shopping preferences.]	1	1.43
To what extent do you agree with the following statements? [I am more likely to purchase from a retailer that offers personalised recommendations.]	1	1.43
To what extent do you agree with the following statements? [I regularly receive personalised recommendations from fashion retailers.]	1	1.43
To what extent do social media platforms or influencers influence your fashion purchasing decisions?	1	1.43
To what extent do discounts influence your fashion purchasing decisions?	1	1.43

What is the single most important factor influencing your fashion purchases?	1	1.43
Which fashion brand do you purchase from most frequently?	1	1.43
Which shopping channel do you use most frequently?	1	1.43
Which fashion product category do you purchase most frequently?	1	1.43
How many fashion purchases do you typically make in a month?	1	1.43
Approximately how much do you spend on fashion products each month?	1	1.43
What is your monthly disposable income?	1	1.43
What is your gender?	1	1.43
What is your age?	1	1.43

5. Clean the dataset

Only respondents who agreed to participate are retained. Administrative fields such as timestamp, consent and email are removed from the analytical dataset.

No artificial zero values are inserted.

In [99]:
```
consent_col = 'Do you consent to participate in this research study?

df_clean = df[
    df[consent_col] == 'I agree to participate in this research stud
].copy()

open_text = [
    'In your opinion, what could fashion retailers do to improve you
    'Is there anything else you would like to share about your fash
]

for col in open_text:
    df_clean[col] = df_clean[col].fillna('No response')

df_clean = df_clean.drop(
    columns=['Timestamp', consent_col, 'Email address'],
    errors='ignore'
)

df_clean = df_clean.drop_duplicates().reset_index(drop=True)

print("Original respondents:", len(df))
print("Consenting respondents:", len(df_clean))
print("Remaining missing cells:", df_clean.isna().sum().sum())
print("Duplicate rows after cleaning:", df_clean.duplicated().sum())
```

```
Original respondents: 70
Consenting respondents: 68
Remaining missing cells: 0
Duplicate rows after cleaning: 0
```

6. Convert ordinal survey ranges

Income, monthly fashion spending and purchase frequency were collected as categories. Approximate midpoints are used to create numerical variables suitable for correlation and regression.

These are analytical approximations and should be acknowledged as a limitation.

In [100]:

```
income_map = {
    'Less than £1,000': 500,
    '£1,000–£1,999': 1500,
    '£2,000–£2,999': 2500,
    '£3,000–£3,999': 3500,
    '£4,000–£4,999': 4500,
    '£5,000 or above': 5500
}

spending_map = {
    'Less than £50': 25,
    '£50–£99': 75,
    '£100–£199': 150,
    '£200–£299': 250,
    '£300–£499': 400,
    '£500 or above': 500
}

purchase_map = {
    '1–2': 1.5,
    '3–4': 3.5,
    '5–6': 5.5,
    '7 or more': 7
}

df_clean['Income_Numeric'] = df_clean[
    'What is your monthly disposable income?'
].map(income_map)

df_clean['Fashion_Spending'] = df_clean[
    'Approximately how much do you spend on fashion products each m
].map(spending_map)

df_clean['Purchase_Frequency'] = df_clean[
    'How many fashion purchases do you typically make in a month?'
].map(purchase_map)

print("Unmapped income:", df_clean['Income_Numeric'].isna().sum())
print("Unmapped spending:", df_clean['Fashion_Spending'].isna().sum
print("Unmapped purchase frequency:", df_clean['Purchase_Frequency'
```

```
Unmapped income: 0
Unmapped spending: 0
Unmapped purchase frequency: 0
```

7. Likert-scale coding and composite variables

The five personalisation questions and five brand-reputation questions use a five-point Likert scale.

- Strongly Disagree = 1
- Disagree = 2
- Neutral = 3
- Agree = 4
- Strongly Agree = 5

The mean of the five items creates each composite score.

In [101]:
```
rt_map = {
'Strongly Disagree': 1,
'Disagree': 2,
'Neutral': 3,
'Agree': 4,
'Strongly Agree': 5


onalisation_cols = [
'To what extent do you agree with the following statements? [I regul
'To what extent do you agree with the following statements? [Persona
'To what extent do you agree with the following statements? [I am mo
'To what extent do you agree with the following statements? [I feel
'To what extent do you agree with the following statements? [Persona


d_cols = [
"Please indicate your level of agreement with the following statemen
"Please indicate your level of agreement with the following statemen
"Please indicate your level of agreement with the following statemen
"Please indicate your level of agreement with the following statemen
"Please indicate your level of agreement with the following statemen


col in personalisation_cols + brand_cols:
df_clean[col + '_Score'] = df_clean[col].map(likert_map)

lean['Personalisation_Score'] = df_clean[
[c + '_Score' for c in personalisation_cols]
an(axis=1)

lean['Brand_Reputation_Score'] = df_clean[
[c + '_Score' for c in brand_cols]
an(axis=1)

lean['Discount_Influence'] = pd.to_numeric(
df_clean['To what extent do discounts influence your fashion purchas
errors='coerce'


lean['Social_Media_Influence'] = pd.to_numeric(
```

```
df_clean['To what extent do social media platforms or influencers in
errors='coerce'


ysis_cols = [
'Income_Numeric',
'Fashion_Spending',
'Purchase_Frequency',
'Discount_Influence',
'Social_Media_Influence',
'Personalisation_Score',
'Brand_Reputation_Score'


lay(df_clean[analysis_cols].isna().sum().to_frame('Missing values'))
```

	Missing values
Income_Numeric	0
Fashion_Spending	0
Purchase_Frequency	0
Discount_Influence	0
Social_Media_Influence	0
Personalisation_Score	0
Brand_Reputation_Score	0

8. Final analytical dataset

Short variable names are used for the statistical models to avoid errors caused by long questionnaire column names.

In [102]:
```
analysis_df = df_clean[analysis_cols].copy()

analysis_df['Age_Group'] = df_clean['What is your age?'].values
analysis_df['Gender'] = df_clean['What is your gender?'].values
analysis_df['Brand'] = df_clean[
    'Which fashion brand do you purchase from most frequently?'
].values
analysis_df['Purchase_Factor'] = df_clean[
    'What is the single most important factor influencing your fash
].values

# Complete-case check for the analytical variables
analysis_df = analysis_df.replace([np.inf, -np.inf], np.nan).dropna

print("Final analytical sample:", len(analysis_df))
display(analysis_df.head())
```

```
Final analytical sample: 68
```

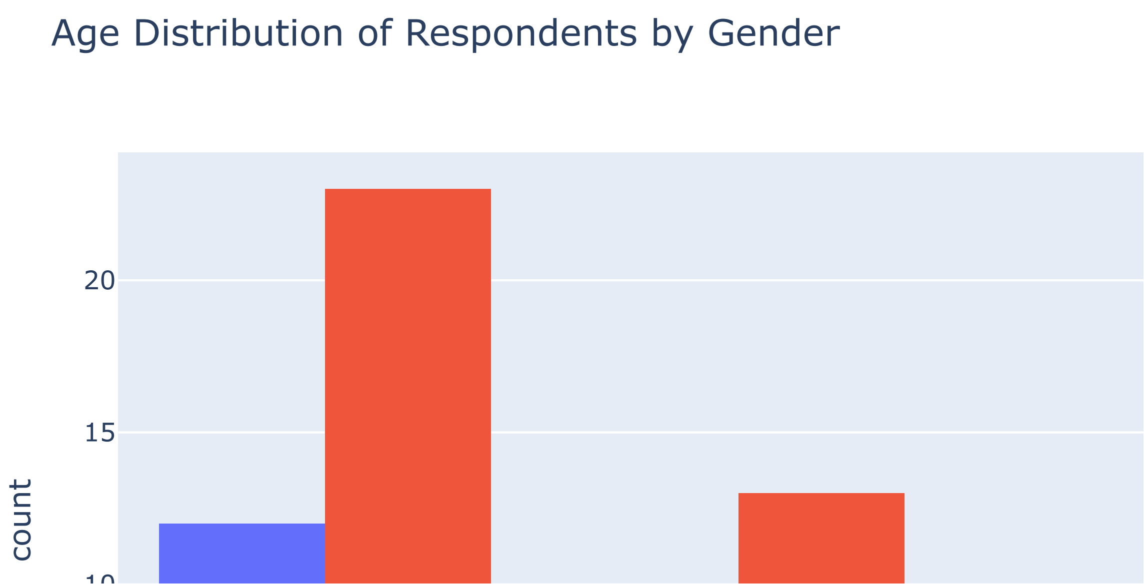
	Income_Numeric	Fashion_Spending	Purchase_Frequency	Discount_Influence	Social_Me
0	1500	75	1.5	3.0	
1	2500	250	3.5	2.0	
2	500	25	3.5	3.0	
3	500	75	1.5	4.0	
4	2500	75	3.5	4.0	

Part A — Interactive Plotly visualisations

Plotly Express is used instead of Seaborn for the main visualisations. Animation is only used across meaningful age groups because this is a cross-sectional survey and does not contain a time series.

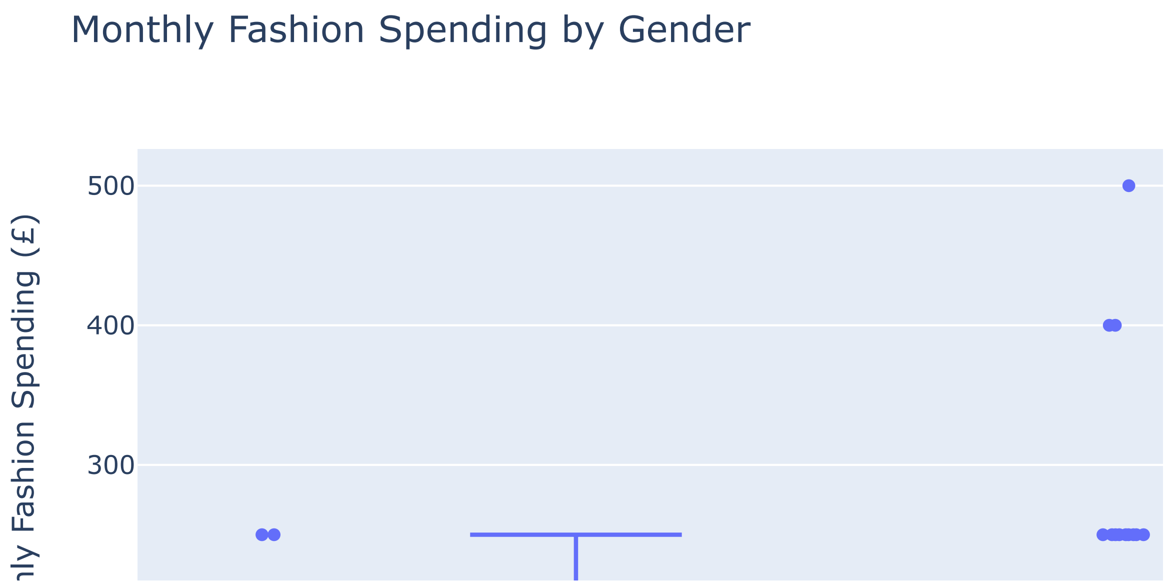
9. Age distribution by gender

```
In [103]: fig = px.histogram(
              analysis_df,
              x='Age_Group',
              color='Gender',
              barmode='group',
              title='Age Distribution of Respondents by Gender',
              labels={'Age_Group':'Age Group','count':'Number of Respondents'
          )
          fig.show()
```



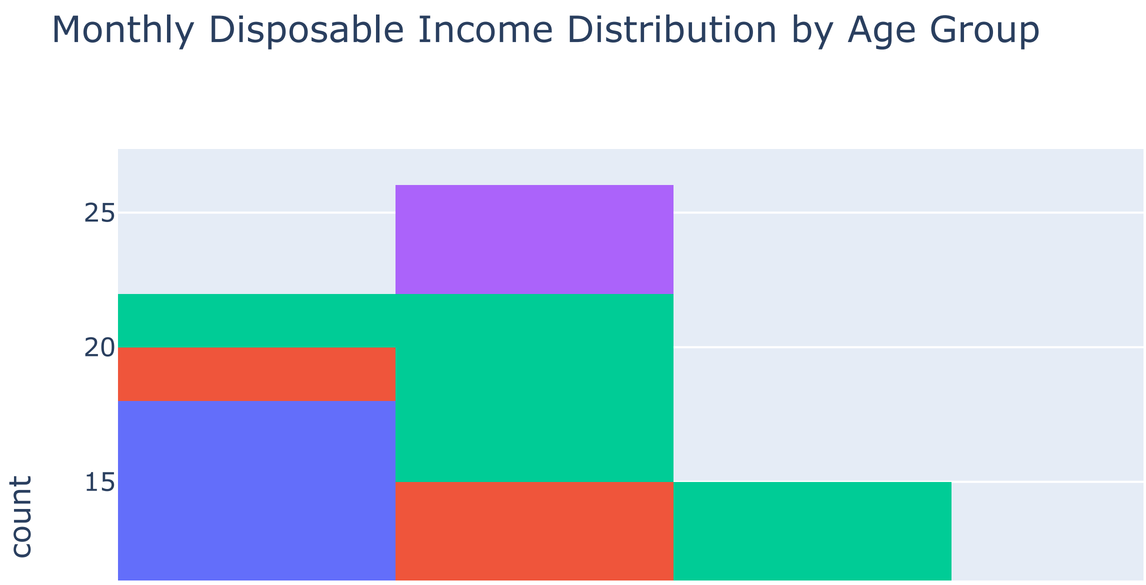
10. Monthly fashion spending by gender

In [104]:
```
fig = px.box(
    analysis_df,
    x='Gender',
    y='Fashion_Spending',
    points='all',
    title='Monthly Fashion Spending by Gender',
    labels={
        'Gender':'Gender',
        'Fashion_Spending':'Estimated Monthly Fashion Spending (£)'
    }
)
fig.show()
```



11. Disposable income distribution

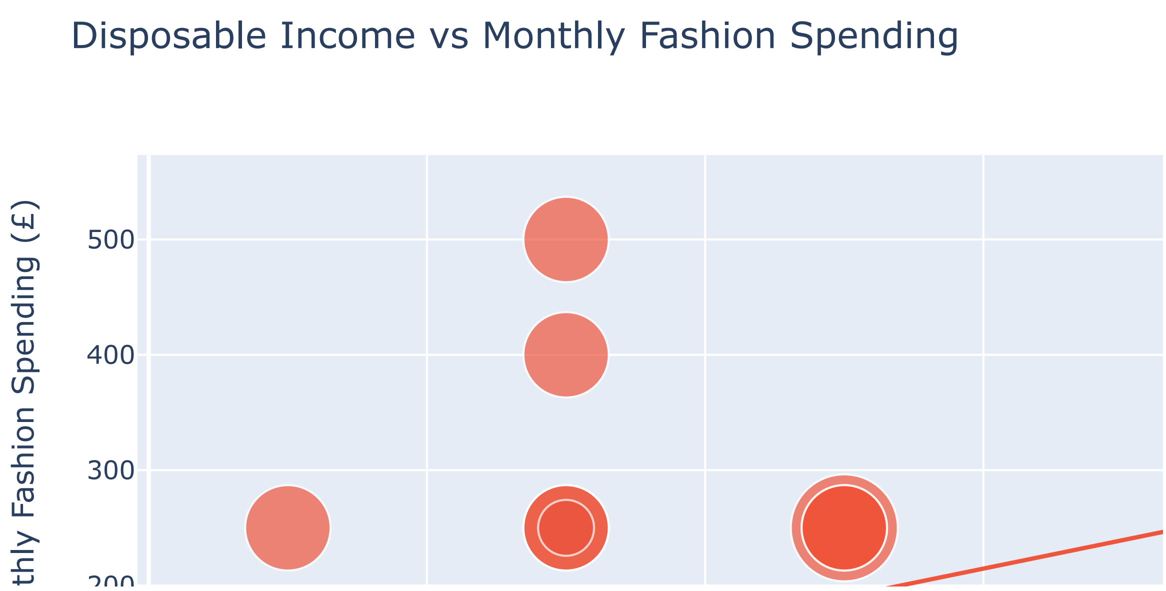
In [105]:
```
fig = px.histogram(
    analysis_df,
    x='Income_Numeric',
    color='Age_Group',
    nbins=6,
    title='Monthly Disposable Income Distribution by Age Group',
    labels={
        'Income_Numeric':'Approximate Monthly Disposable Income (£)
        'count':'Respondents'
    }
)
fig.show()
```



12. Income versus fashion spending

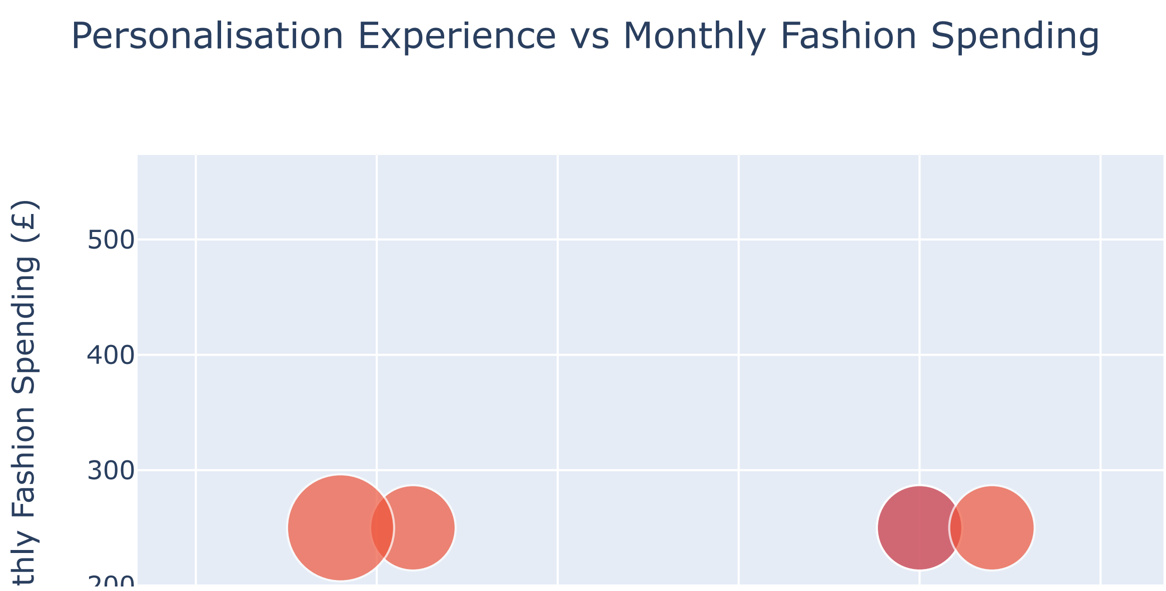
The OLS trendline provides a visual representation of the relationship later tested formally using Pearson correlation and simple linear regression.

```
In [106]: fig = px.scatter(
    analysis_df,
    x='Income_Numeric',
    y='Fashion_Spending',
    color='Gender',
    size='Purchase_Frequency',
    hover_data=['Age_Group','Personalisation_Score','Brand_Reputati
    trendline='ols',
    title='Disposable Income vs Monthly Fashion Spending',
    labels={
        'Income_Numeric':'Approximate Monthly Disposable Income (£)
        'Fashion_Spending':'Approximate Monthly Fashion Spending (£
        'Purchase_Frequency':'Purchase Frequency'
    },
    size_max=40
)
fig.show()
```



13. Personalisation versus fashion spending

```
In [107]: fig = px.scatter(
    analysis_df,
    x='Personalisation_Score',
    y='Fashion_Spending',
    color='Gender',
    size='Purchase_Frequency',
    hover_data=['Age_Group','Income_Numeric','Brand_Reputation_Scor
    trendline='ols',
    title='Personalisation Experience vs Monthly Fashion Spending',
    labels={
        'Personalisation_Score':'Personalisation Score (1–5)',
        'Fashion_Spending':'Approximate Monthly Fashion Spending (£
        'Purchase_Frequency':'Purchase Frequency'
    },
    size_max=40
)
fig.show()
```

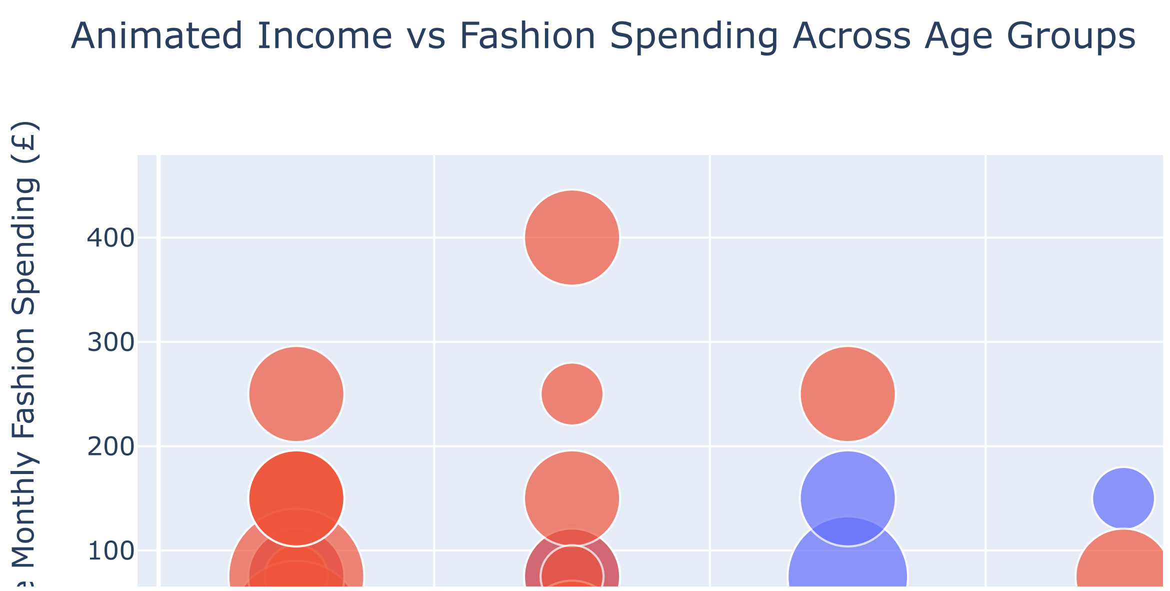


14. Animated income and spending by age group

This is the closest equivalent to the animated Plotly examples supplied. The animation compares age groups rather than pretending that the survey contains observations over time.

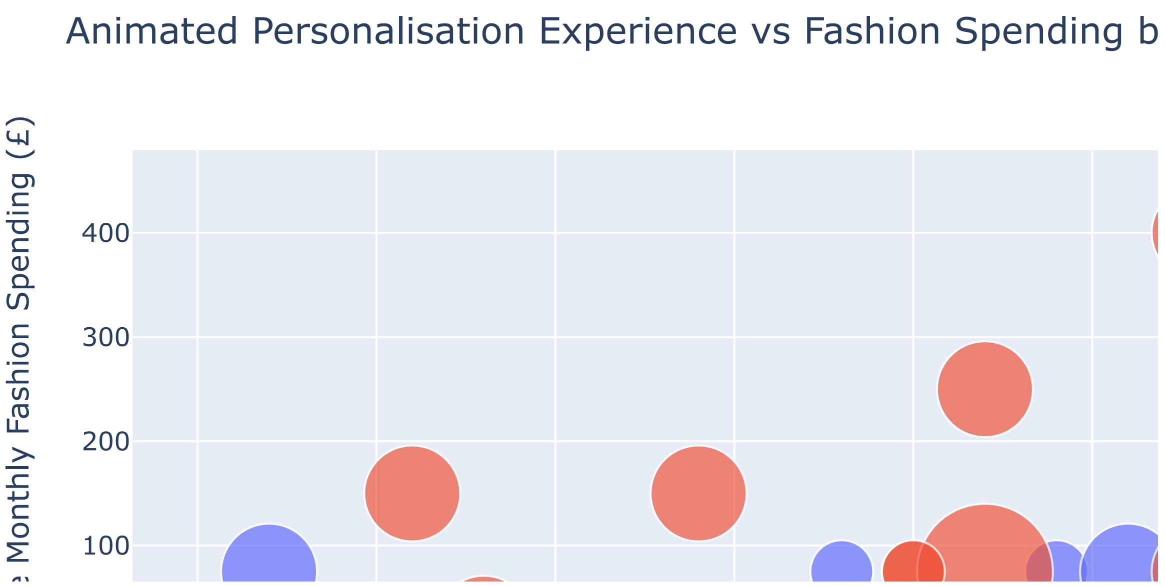
In [108]:

```
fig = px.scatter(
    analysis_df,
    x='Income_Numeric',
    y='Fashion_Spending',
    animation_frame='Age_Group',
    color='Gender',
    size='Purchase_Frequency',
    hover_data=['Personalisation_Score','Brand_Reputation_Score'],
    title='Animated Income vs Fashion Spending Across Age Groups',
    labels={
        'Income_Numeric':'Approximate Monthly Disposable Income (£)
        'Fashion_Spending':'Approximate Monthly Fashion Spending (£
        'Purchase_Frequency':'Purchase Frequency'
    },
    size_max=45
)
fig.show()
```



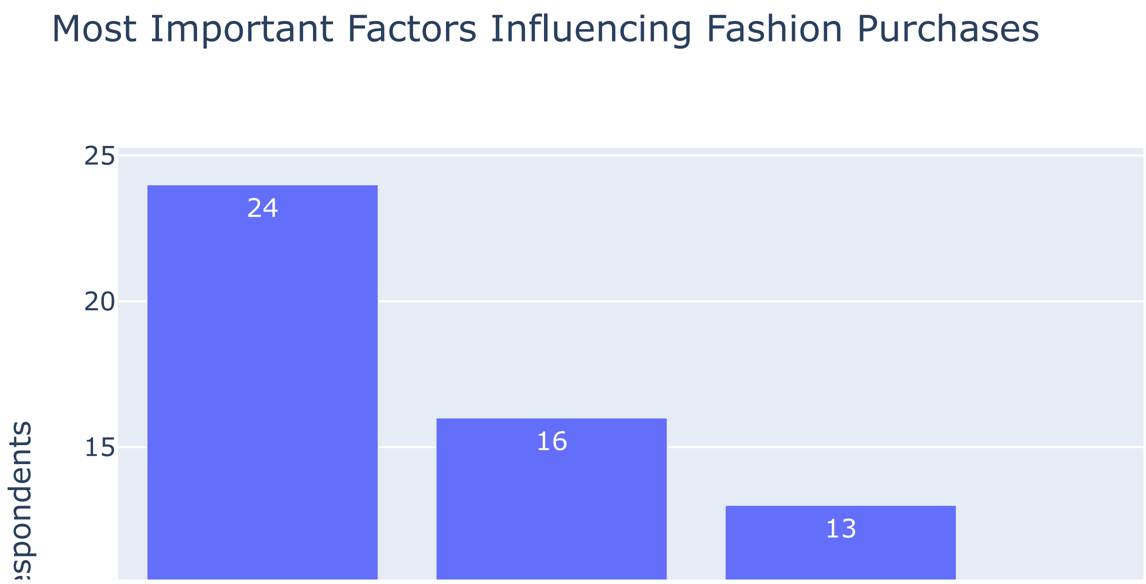
15. Animated personalisation and spending by age group

In [109]:
```
fig = px.scatter(
    analysis_df,
    x='Personalisation_Score',
    y='Fashion_Spending',
    animation_frame='Age_Group',
    color='Gender',
    size='Purchase_Frequency',
    hover_data=['Income_Numeric','Brand_Reputation_Score'],
    title='Animated Personalisation Experience vs Fashion Spending
    labels={
        'Personalisation_Score':'Personalisation Score (1–5)',
        'Fashion_Spending':'Approximate Monthly Fashion Spending (£
        'Purchase_Frequency':'Purchase Frequency'
    },
    size_max=45
)
fig.show()
```



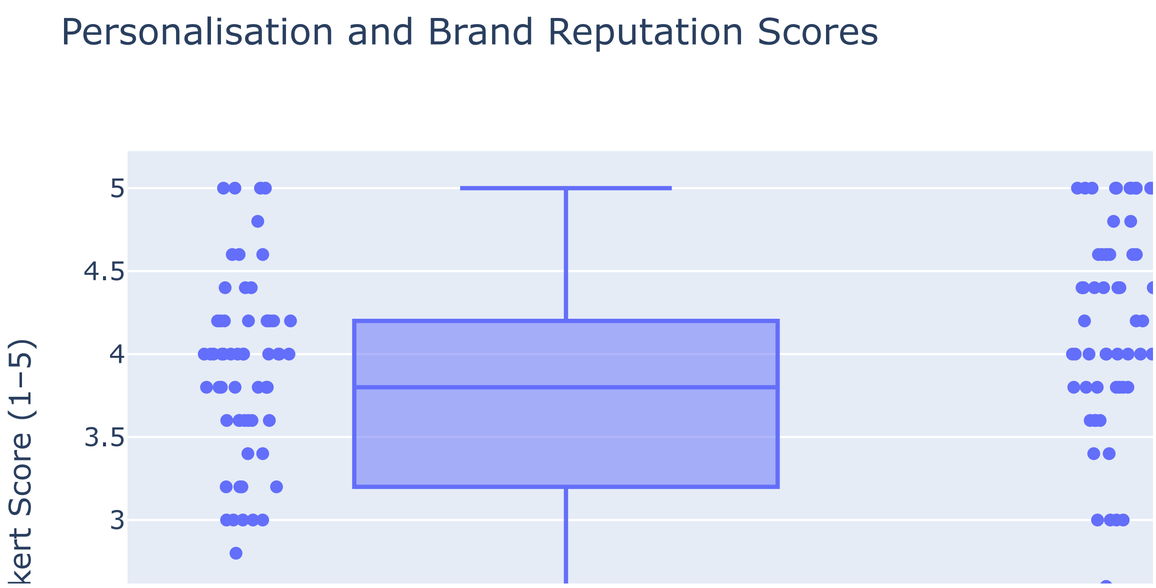
16. Most important purchase factors

In [110]:
```
factor_counts = (
    analysis_df['Purchase_Factor']
    .value_counts()
    .rename_axis('Purchase_Factor')
    .reset_index(name='Respondents')
)

fig = px.bar(
    factor_counts,
    x='Purchase_Factor',
    y='Respondents',
    text='Respondents',
    title='Most Important Factors Influencing Fashion Purchases',
    labels={'Purchase_Factor':'Purchase Factor','Respondents':'Resp
)
fig.show()
```



17. Personalisation and brand-reputation scores

In [111]:
```
score_long = analysis_df[
    ['Personalisation_Score','Brand_Reputation_Score']
].melt(var_name='Construct', value_name='Score')

fig = px.box(
    score_long,
    x='Construct',
    y='Score',
    points='all',
    title='Personalisation and Brand Reputation Scores',
    labels={'Construct':'Construct','Score':'Mean Likert Score (1–5
)
fig.show()
```



Part B — Descriptive statistics and correlation

Descriptive statistics establish the central tendency and spread of the main analytical variables before inferential testing.

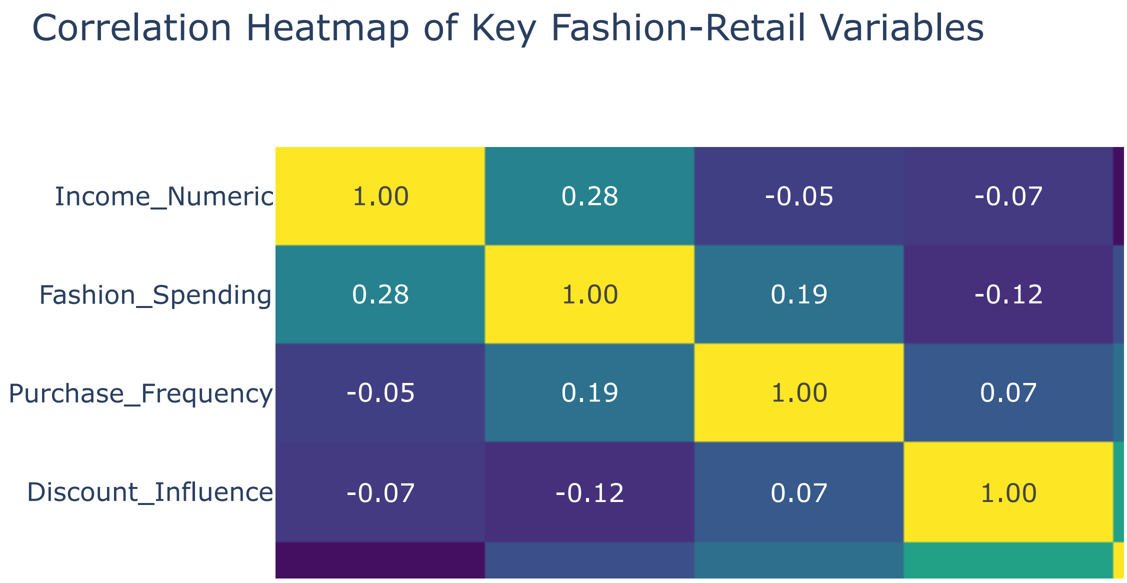
In [112]:
```
descriptive = analysis_df[analysis_cols].describe().T.round(3)
display(descriptive)
```

	count	mean	std	min	25%	50%	75%	max
Income_Numeric	68.0	1588.235	1047.175	500.0	500.0	1500.0	2500.0	5500.0
Fashion_Spending	68.0	126.103	99.854	25.0	75.0	75.0	150.0	500.0
Purchase_Frequency	68.0	2.890	1.401	1.5	1.5	3.5	3.5	7.0
Discount_Influence	68.0	3.456	1.177	1.0	3.0	3.5	4.0	5.0
Social_Media_Influence	68.0	3.118	1.179	1.0	2.0	3.0	4.0	5.0
Personalisation_Score	68.0	3.621	0.920	1.2	3.2	3.8	4.2	5.0
Brand_Reputation_Score	68.0	3.926	1.043	1.0	3.6	4.0	4.6	5.0

18. Correlation matrix and interactive heatmap

Pearson correlation is used for the numerical analytical variables. The heatmap provides a compact visual summary.

In [113]:
```
corr_matrix = analysis_df[analysis_cols].corr()

fig = px.imshow(
    corr_matrix,
    text_auto='.2f',
    aspect='auto',
    color_continuous_scale='Viridis',
    title='Correlation Heatmap of Key Fashion-Retail Variables',
    labels={'color':'Correlation'}
)
fig.show()
```



19. Pearson correlation — income and fashion spending

H1: Disposable income is significantly associated with monthly fashion spending.

H0: Disposable income is not significantly associated with monthly fashion spending.

In [114]:
```
r_income, p_income = stats.pearsonr(
    analysis_df['Income_Numeric'],
    analysis_df['Fashion_Spending']
)

print(f"Correlation (r): {r_income:.3f}")
print(f"P-value: {p_income:.4f}")
print(f"N: {len(analysis_df)}")
print(
    "Decision:",
    "Reject H0 at 5% significance."
    if p_income < 0.05
    else "Fail to reject H0 at 5% significance."
)
```

```
Correlation (r): 0.285
P-value: 0.0187
N: 68
Decision: Reject H0 at 5% significance.
```

20. Pearson correlation — personalisation and fashion spending

H2: Personalisation experience is significantly associated with monthly fashion spending.

H0: Personalisation experience is not significantly associated with monthly fashion spending.

In [115]:
```
r_pers, p_pers = stats.pearsonr(
    analysis_df['Personalisation_Score'],
    analysis_df['Fashion_Spending']
)

print(f"Correlation (r): {r_pers:.3f}")
print(f"P-value: {p_pers:.4f}")
print(f"N: {len(analysis_df)}")
print(
    "Decision:",
    "Reject H0 at 5% significance."
    if p_pers < 0.05
    else "Fail to reject H0 at 5% significance."
)
```

```
Correlation (r): 0.016
P-value: 0.8969
N: 68
Decision: Fail to reject H0 at 5% significance.
```

Part C — Simple linear regression

The reference methodology uses simple OLS regression before moving to multiple regression. Two simple models are used here: income and personalisation as predictors of monthly fashion spending.

21. Simple linear regression — Income

In [116]:
```
model1 = ols(
    'Fashion_Spending ~ Income_Numeric',
    data=analysis_df
).fit()

print(model1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       Fashion_Spending   R-squared:                       0.081
Model:                            OLS   Adj. R-squared:                  0.067
Method:                 Least Squares   F-statistic:                     5.814
Date:                Sun, 23 Aug 2026   Prob (F-statistic):             0.0187
Time:                        01:43:13   Log-Likelihood:                -406.17
No. Observations:                  68   AIC:                             816.3
Df Residuals:                      66   BIC:                             820.8
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept         83.0114     21.358      3.887      0.000      40.368     125.655
Income_Numeric     0.0271      0.011      2.411      0.019       0.005       0.050
==============================================================================
Omnibus:                       25.375   Durbin-Watson:                   2.593
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               43.577
Skew:                           1.347   Prob(JB):                     3.45e-10
Kurtosis:                       5.850   Cond. No.                     3.47e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.47e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

22. Simple regression visualisation — Income

In [117]:
```
fig = px.scatter(
    analysis_df,
    x='Income_Numeric',
    y='Fashion_Spending',
    color='Gender',
    trendline='ols',
    title='Simple Linear Regression: Income and Fashion Spending',
    labels={
        'Income_Numeric':'Approximate Monthly Disposable Income (£)
        'Fashion_Spending':'Approximate Monthly Fashion Spending (£
    }
)
fig.show()
```

23. Simple linear regression — Personalisation

In [118]:
```python
model2 = ols(
    'Fashion_Spending ~ Personalisation_Score',
    data=analysis_df
).fit()

print(model2.summary())
```

```
                            OLS Regression Results
==================================================================
============
Dep. Variable:       Fashion_Spending   R-squared:
0.000
Model:                            OLS   Adj. R-squared:
-0.015
Method:                 Least Squares   F-statistic:
0.01691
Date:                Sun, 23 Aug 2026   Prob (F-statistic):
0.897
Time:                        01:43:13   Log-Likelihood:
-409.03
No. Observations:                  68   AIC:
822.1
Df Residuals:                      66   BIC:
826.5
Df Model:                           1
Covariance Type:            nonrobust
==================================================================
=======================
                            coef    std err          t      P>|t|
 [0.025      0.975]
------------------------------------------------------------------
-----------------------
Intercept               119.8106     49.903      2.401      0.019
20.176     219.446
Personalisation_Score     1.7379     13.365      0.130      0.897
-24.946      28.422
==================================================================
============
Omnibus:                       25.497   Durbin-Watson:
2.583
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
38.967
Skew:                           1.464   Prob(JB):
3.45e-09
Kurtosis:                       5.275   Cond. No.
16.3
==================================================================
============

Notes:
[1] Standard Errors assume that the covariance matrix of the error
s is correctly specified.
```

24. Simple regression visualisation — Personalisation

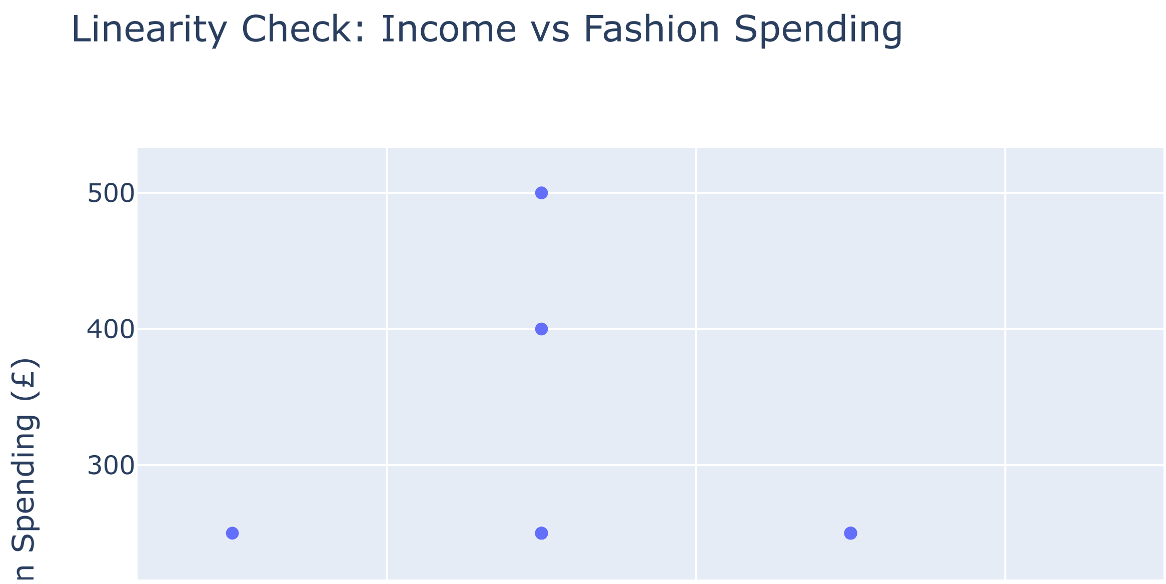
In [119]:
```
fig = px.scatter(
    analysis_df,
    x='Personalisation_Score',
    y='Fashion_Spending',
    color='Gender',
    trendline='ols',
    title='Simple Linear Regression: Personalisation and Fashion Sp
    labels={
        'Personalisation_Score':'Personalisation Score (1–5)',
        'Fashion_Spending':'Approximate Monthly Fashion Spending (£
    }
)
fig.show()
```

Part D — OLS assumptions and diagnostics

The following tests and plots assess linearity, homoscedasticity, residual normality and independence. These diagnostics should be interpreted together.

25. Linearity check

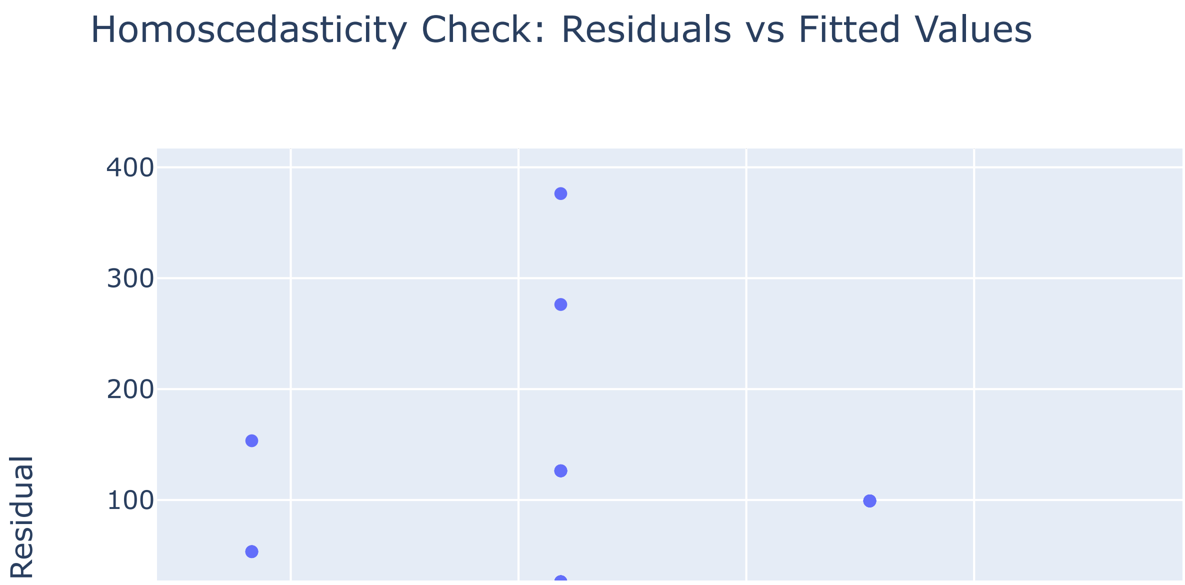
In [120]:
```
fig = px.scatter(
    analysis_df,
    x='Income_Numeric',
    y='Fashion_Spending',
    trendline='ols',
    title='Linearity Check: Income vs Fashion Spending',
    labels={
        'Income_Numeric':'Income (£)',
        'Fashion_Spending':'Fashion Spending (£)'
    }
)
fig.show()
```



26. Homoscedasticity — residuals versus fitted values

A roughly random spread around zero without a strong funnel pattern supports constant residual variance.

In [121]:
```
resid1 = pd.DataFrame({
    'Fitted': model1.fittedvalues,
    'Residual': model1.resid
})

fig = px.scatter(
    resid1,
    x='Fitted',
    y='Residual',
    trendline='lowess',
    title='Homoscedasticity Check: Residuals vs Fitted Values',
    labels={'Fitted':'Fitted Values','Residual':'Residual'}
)
fig.add_hline(y=0, line_dash='dash')
fig.show()
```



27. Residual histogram

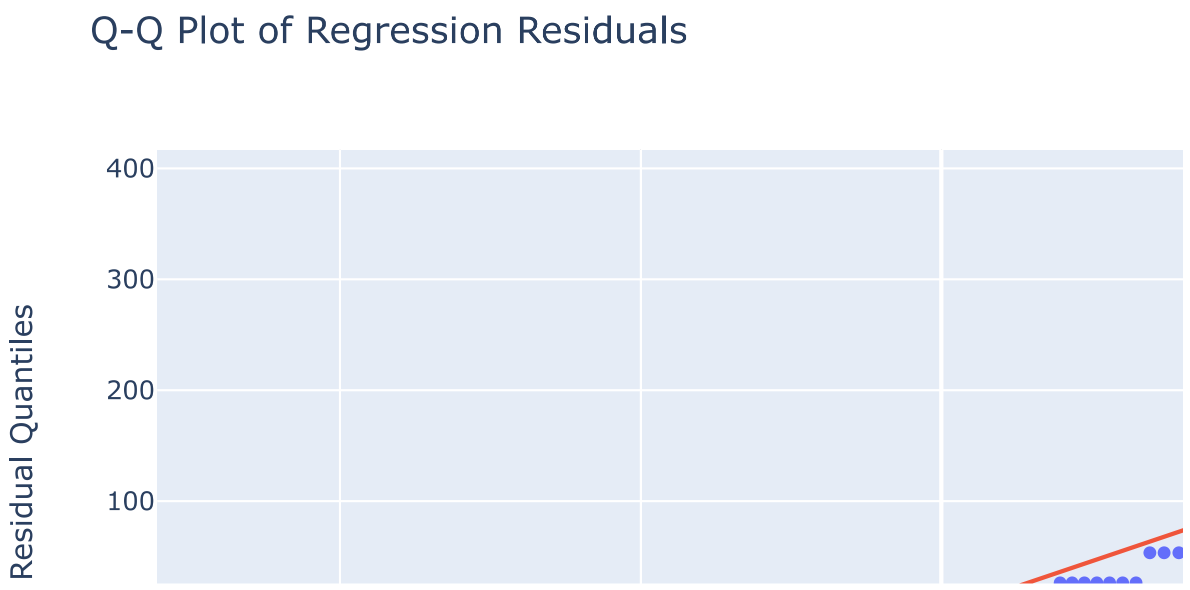
In [122]:
```
fig = px.histogram(
    x=model1.resid,
    nbins=12,
    marginal='box',
    title='Histogram of Simple Regression Residuals',
    labels={'x':'Residual'}
)
fig.show()
```

28. Q-Q plot

The Q-Q plot compares observed residual quantiles with theoretical normal quantiles.

In [123]:
```
theoretical, ordered = stats.probplot(
    model1.resid, dist='norm'
)[0]

slope, intercept = stats.probplot(
    model1.resid, dist='norm'
```

```
)[1][:2]

qq = pd.DataFrame({
    'Theoretical': theoretical,
    'Observed': ordered
})

xline = np.array([
    qq['Theoretical'].min(),
    qq['Theoretical'].max()
])
yline = intercept + slope * xline

fig = px.scatter(
    qq,
    x='Theoretical',
    y='Observed',
    title='Q-Q Plot of Regression Residuals',
    labels={
        'Theoretical':'Theoretical Quantiles',
        'Observed':'Observed Residual Quantiles'
    }
)

fig.add_trace(
    go.Scatter(
        x=xline,
        y=yline,
        mode='lines',
        name='Reference line'
    )
)

fig.show()
```



29. Shapiro-Wilk normality test

A p-value above 0.05 indicates insufficient evidence of non-normality at the 5% level.

In [124]:
```
shapiro1 = stats.shapiro(model1.resid)

print(f"Statistic: {shapiro1.statistic:.4f}")
print(f"P-value: {shapiro1.pvalue:.4f}")

if shapiro1.pvalue >= 0.05:
    print("Conclusion: No statistically significant departure from
else:
    print("Conclusion: Evidence of non-normal residuals at 5%; insp
```

```
Statistic: 0.8894
P-value: 0.0000
Conclusion: Evidence of non-normal residuals at 5%; inspect the Q-
Q plot.
```

30. Durbin-Watson test

A statistic close to 2 suggests little residual autocorrelation.

In [125]:
```
dw1 = durbin_watson(model1.resid)
print(f"Durbin-Watson statistic: {dw1:.3f}")
```

```
Durbin-Watson statistic: 2.593
```

Part E — Multiple linear regression

The multiple-regression models are built progressively, similar to the staged modelling approach in the supplied reference notebook. Monthly fashion spending remains the dependent variable.

31. Model 1 — Basic behaviour

Predictors: purchase frequency and discount influence.

In [126]:
```
model_m1 = ols(
    'Fashion_Spending ~ Purchase_Frequency + Discount_Influence',
    data=analysis_df
).fit()

print(model_m1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       Fashion_Spending   R-squared:                       0.052
Model:                            OLS   Adj. R-squared:                  0.023
Method:                 Least Squares   F-statistic:                     1.798
Date:                Sun, 23 Aug 2026   Prob (F-statistic):              0.174
Time:                        01:43:13   Log-Likelihood:                -407.20
No. Observations:                  68   AIC:                             820.4
Df Residuals:                      65   BIC:                             827.1
Df Model:                           2
Covariance Type:            nonrobust
======================================================================================
                         coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------
Intercept            123.2525     43.557      2.830      0.006      36.263     210.242
Purchase_Frequency    14.1364      8.630      1.638      0.106      -3.100      31.373
Discount_Influence   -10.9956     10.267     -1.071      0.288     -31.500       9.509
==============================================================================
Omnibus:                       19.185   Durbin-Watson:                   2.540
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               24.711
Skew:                           1.207   Prob(JB):                     4.31e-06
Kurtosis:                       4.700   Cond. No.                         17.8
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

32. Model 2 — Add brand reputation

In [127]:
```
model_m2 = ols(
    'Fashion_Spending ~ Purchase_Frequency + Discount_Influence + B
    data=analysis_df
).fit()

print(model_m2.summary())
```

```
                            OLS Regression Results
==================================================================
============
Dep. Variable:       Fashion_Spending   R-squared:
0.062
Model:                            OLS   Adj. R-squared:
0.018
Method:                 Least Squares   F-statistic:
1.416
Date:                Sun, 23 Aug 2026   Prob (F-statistic):
0.246
Time:                        01:43:13   Log-Likelihood:
-406.85
No. Observations:                  68   AIC:
821.7
Df Residuals:                      64   BIC:
830.6
Df Model:                           3
Covariance Type:            nonrobust
==================================================================
========================
                            coef    std err          t      P>|t|
[0.025      0.975]
------------------------------------------------------------------
------------------------
Intercept                76.8153     71.614      1.073      0.287
-66.249     219.880
Purchase_Frequency       16.2678      9.036      1.800      0.077
-1.784      34.320
Discount_Influence      -10.6092     10.304     -1.030      0.307
-31.194       9.975
Brand_Reputation_Score    9.9179     12.122      0.818      0.416
-14.299      34.135
==================================================================
============
Omnibus:                       18.248   Durbin-Watson:
2.486
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
22.871
Skew:                           1.170   Prob(JB):
1.08e-05
Kurtosis:                       4.611   Cond. No.
37.2
==================================================================
============
```

```
Notes:
[1] Standard Errors assume that the covariance matrix of the error
s is correctly specified.
```

33. Model 3 — Add social-media influence

In [128]:
```
model_m3 = ols(
    'Fashion_Spending ~ Purchase_Frequency + Discount_Influence + B
    data=analysis_df
).fit()

print(model_m3.summary())
```

```
                            OLS Regression Results
==============================================================================
============
Dep. Variable:       Fashion_Spending   R-squared:
0.065
Model:                            OLS   Adj. R-squared:
0.005
Method:                 Least Squares   F-statistic:
1.091
Date:                Sun, 23 Aug 2026   Prob (F-statistic):
0.369
Time:                        01:43:13   Log-Likelihood:
-406.76
No. Observations:                  68   AIC:
823.5
Df Residuals:                      63   BIC:
834.6
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
========================
                            coef    std err          t      P>|t|
[0.025      0.975]
------------------------------------------------------------------------------
------------------------
Intercept                72.2353     72.921      0.991      0.326
-73.487     217.957
Purchase_Frequency       15.6074      9.233      1.690      0.096
-2.844      34.059
Discount_Influence      -12.7342     11.567     -1.101      0.275
-35.848      10.380
Brand_Reputation_Score    9.5852     12.228      0.784      0.436
-14.850      34.021
Social_Media_Influence    4.8558     11.703      0.415      0.680
-18.530      28.242
==============================================================================
============
Omnibus:                       18.343   Durbin-Watson:
2.473
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
23.160
Skew:                           1.168   Prob(JB):
```

```
9.35e-06
Kurtosis:                       4.649   Cond. No.
42.3
==================================================================
============

Notes:
[1] Standard Errors assume that the covariance matrix of the error
s is correctly specified.
```

34. Model 4 — Full dissertation model

The final model includes income, purchase frequency, discount influence, social-media influence, personalisation and brand reputation.

In [129]:

```
ome_Numeric + Purchase_Frequency + Discount_Influence + Social_Media
```

```
                            OLS Regression Results
==================================================================
============
Dep. Variable:      Fashion_Spending   R-squared:
0.154
Model:                           OLS   Adj. R-squared:
0.071
Method:                Least Squares   F-statistic:
1.855
Date:               Sun, 23 Aug 2026   Prob (F-statistic):
0.103
Time:                       01:43:13   Log-Likelihood:
-403.34
No. Observations:                 68   AIC:
820.7
Df Residuals:                     61   BIC:
836.2
Df Model:                          6
Covariance Type:           nonrobust
==================================================================
========================
                         coef    std err          t      P>|t|
[0.025      0.975]
------------------------------------------------------------------
------------------------
Intercept             25.3067     73.742      0.343      0.733
-122.149     172.762
Income_Numeric         0.0294      0.012      2.526      0.014
0.006        0.053
Purchase_Frequency    15.1681      8.971      1.691      0.096
-2.770       33.107
Discount_Influence   -13.9760     11.242     -1.243      0.219
-36.455        8.503
```

```
Social_Media_Influence    12.3106     11.993      1.027      0.309
-11.670     36.291
Personalisation_Score    -2.9044     17.077     -0.170      0.866
-37.052     31.243
Brand_Reputation_Score    7.8039     15.206      0.513      0.610
-22.602     38.210
==================================================================
============
Omnibus:                       21.527   Durbin-Watson:
2.476
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
35.288
Skew:                           1.151   Prob(JB):
2.17e-08
Kurtosis:                       5.675   Cond. No.
1.21e+04
==================================================================
============

Notes:
[1] Standard Errors assume that the covariance matrix of the error
s is correctly specified.
[2] The condition number is large, 1.21e+04. This might indicate t
hat there are
strong multicollinearity or other numerical problems.
```

35. Model comparison

R^2 measures explained variance. Adjusted R^2 accounts for model complexity. AIC balances model fit and complexity, with lower values generally preferred when models use the same sample and dependent variable.

In [130]:
```
models = {
    'Model 1: Behaviour': model_m1,
    'Model 2: + Brand Reputation': model_m2,
    'Model 3: + Social Media': model_m3,
    'Model 4: Full Model': model_m4
}

comparison = pd.DataFrame({
    'Model': list(models.keys()),
    'R_squared': [m.rsquared for m in models.values()],
    'Adjusted_R_squared': [m.rsquared_adj for m in models.values()]
    'AIC': [m.aic for m in models.values()],
    'BIC': [m.bic for m in models.values()],
    'Model_p_value': [m.f_pvalue for m in models.values()]
})

display(comparison.round(4))
```

	Model	R_squared	Adjusted_R_squared	AIC	BIC	Model_p_value
0	Model 1: Behaviour	0.0524	0.0233	820.4098	827.0684	0.1737
1	Model 2: + Brand Reputation	0.0622	0.0183	821.7023	830.5804	0.2463
2	Model 3: + Social Media	0.0648	0.0054	823.5167	834.6143	0.3686
3	Model 4: Full Model	0.1543	0.0711	820.6749	836.2115	0.1033

36. Interactive model comparison

In [131]:
```
fig = px.bar(
    comparison,
    x='Model',
    y=['R_squared','Adjusted_R_squared'],
    barmode='group',
    title='Comparison of Multiple Regression Models',
    labels={'value':'Model Fit','variable':'Metric'}
)
fig.show()
```

Part F — Full-model diagnostics

The final model is the main inferential model, so its residuals, normality and multicollinearity are checked before interpreting the coefficients.

37. Full-model residual plot

```
In [132]: full_resid = pd.DataFrame({
              'Fitted': model_m4.fittedvalues,
              'Residual': model_m4.resid
          })

          fig = px.scatter(
              full_resid,
              x='Fitted',
              y='Residual',
              trendline='lowess',
              title='Full Model: Residuals vs Fitted Values',
              labels={'Fitted':'Fitted Fashion Spending (£)','Residual':'Resi
          )
          fig.add_hline(y=0, line_dash='dash')
          fig.show()
```

38. Full-model residual histogram

```
In [133]: fig = px.histogram(
              x=model_m4.resid,
              nbins=12,
              marginal='box',
              title='Full Multiple Regression Residual Distribution',
              labels={'x':'Residual'}
          )
          fig.show()
```

39. Full-model Q-Q plot

```
In [134]: theoretical, ordered = stats.probplot(model_m4.resid, dist='norm')[
          slope, intercept = stats.probplot(model_m4.resid, dist='norm')[1][:

          qq = pd.DataFrame({
              'Theoretical': theoretical,
              'Observed': ordered
          })

          xline = np.array([qq['Theoretical'].min(), qq['Theoretical'].max()]
          yline = intercept + slope * xline

          fig = px.scatter(
```

```
    qq,
    x='Theoretical',
    y='Observed',
    title='Q-Q Plot: Full Multiple Regression Residuals',
    labels={'Theoretical':'Theoretical Quantiles','Observed':'Obser
)

fig.add_trace(
    go.Scatter(
        x=xline,
        y=yline,
        mode='lines',
        name='Reference line'
    )
)

fig.show()
```

40. Full-model Shapiro-Wilk and Durbin-Watson

In [135]:
```python
shapiro_full = stats.shapiro(model_m4.resid)
dw_full = durbin_watson(model_m4.resid)

print(f"Shapiro-Wilk statistic: {shapiro_full.statistic:.4f}")
print(f"Shapiro-Wilk p-value: {shapiro_full.pvalue:.4f}")
print(f"Durbin-Watson statistic: {dw_full:.3f}")
```

```
Shapiro-Wilk statistic: 0.9283
Shapiro-Wilk p-value: 0.0008
Durbin-Watson statistic: 2.476
```

41. Multicollinearity — Variance Inflation Factor

VIF is included as an additional diagnostic. Values near 1 indicate very low multicollinearity; substantially larger values require investigation.

In [136]:
```python
vif_vars = [
    'Income_Numeric',
    'Purchase_Frequency',
    'Discount_Influence',
    'Social_Media_Influence',
    'Personalisation_Score',
    'Brand_Reputation_Score'
]

X_vif = sm.add_constant(analysis_df[vif_vars])

vif_table = pd.DataFrame({
    'Variable': X_vif.columns,
    'VIF': [
        variance_inflation_factor(X_vif.values, i)
        for i in range(X_vif.shape[1])
    ]
})

display(vif_table.round(3))
```

	Variable	VIF
0	const	39.926
1	Income_Numeric	1.078
2	Purchase_Frequency	1.142
3	Discount_Influence	1.267
4	Social_Media_Influence	1.445
5	Personalisation_Score	1.784
6	Brand_Reputation_Score	1.821

Part G — Hypotheses and final interpretation

The final model is used to assess the proposed relationships while controlling for the other predictors.

42. Full-model coefficient table

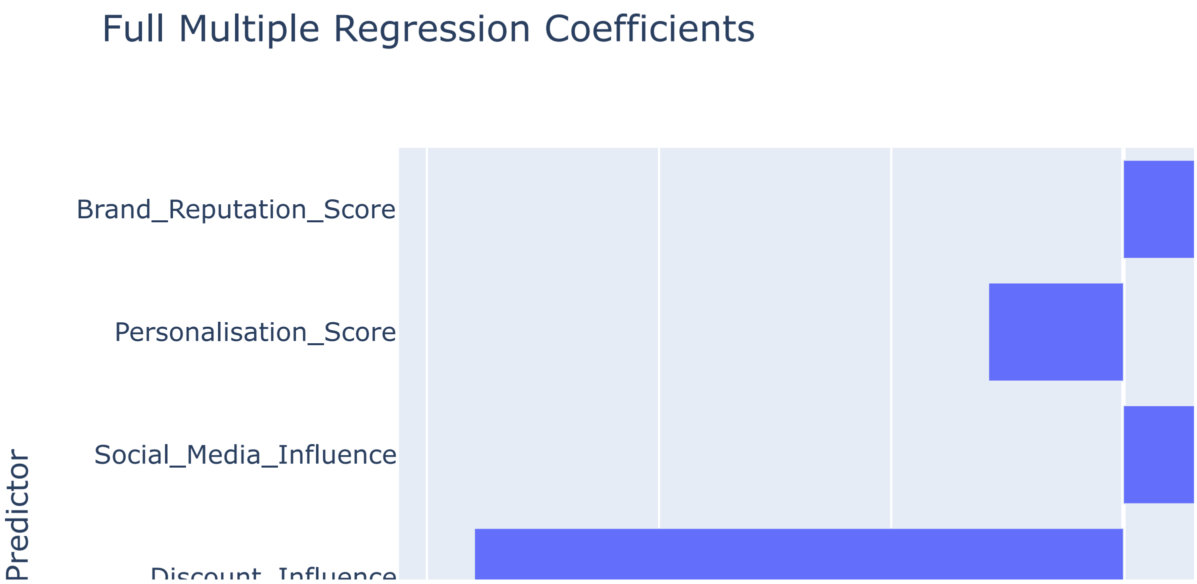
In [137]:
```
coef_table = pd.DataFrame({
    'Coefficient': model_m4.params,
    'Std_Error': model_m4.bse,
    't_value': model_m4.tvalues,
    'p_value': model_m4.pvalues,
    'CI_Lower': model_m4.conf_int()[0],
    'CI_Upper': model_m4.conf_int()[1]
})

display(coef_table.round(4))
```

	Coefficient	Std_Error	t_value	p_value	CI_Lower	CI_Upper
Intercept	25.3067	73.7415	0.3432	0.7326	-122.1486	172.7619
Income_Numeric	0.0294	0.0117	2.5259	0.0142	0.0061	0.0527
Purchase_Frequency	15.1681	8.9709	1.6908	0.0960	-2.7702	33.1065
Discount_Influence	-13.9760	11.2415	-1.2433	0.2185	-36.4548	8.5027
Social_Media_Influence	12.3106	11.9927	1.0265	0.3087	-11.6703	36.2914
Personalisation_Score	-2.9044	17.0771	-0.1701	0.8655	-37.0522	31.2434
Brand_Reputation_Score	7.8039	15.2059	0.5132	0.6097	-22.6021	38.2099

43. Coefficient visualisation

In [138]:

```
coef_plot = coef_table.drop(index='Intercept').reset_index()
coef_plot.columns = [
    'Variable','Coefficient','Std_Error',
    't_value','p_value','CI_Lower','CI_Upper'
]

fig = px.bar(
    coef_plot,
    x='Coefficient',
    y='Variable',
    orientation='h',
    hover_data=['p_value','CI_Lower','CI_Upper'],
    title='Full Multiple Regression Coefficients',
    labels={'Coefficient':'Estimated Coefficient','Variable':'Predi
)
fig.show()
```



44. Hypothesis testing summary

At the 5% significance level:

- $p < 0.05$ → statistically significant association
- $p \geq 0.05$ → insufficient evidence of a statistically significant association

The result should be reported as an association, not a causal effect.

In [139]:
```
hypotheses = {
    'H1: Income → Fashion Spending': 'Income_Numeric',
    'H2: Purchase Frequency → Fashion Spending': 'Purchase_Frequenc
    'H3: Discount Influence → Fashion Spending': 'Discount_Influenc
    'H4: Social Media Influence → Fashion Spending': 'Social_Media_
    'H5: Personalisation → Fashion Spending': 'Personalisation_Scor
    'H6: Brand Reputation → Fashion Spending': 'Brand_Reputation_Sc
}

rows = []

for hypothesis, variable in hypotheses.items():
    p = model_m4.pvalues[variable]
    coef = model_m4.params[variable]

    rows.append({
        'Hypothesis': hypothesis,
        'Coefficient': coef,
        'P-value': p,
        'Decision': (
            'Supported at 5%'
            if p < 0.05
            else 'Not supported at 5%'
        )
    })

hypothesis_table = pd.DataFrame(rows)
display(hypothesis_table.round(4))
```

	Hypothesis	Coefficient	P-value	Decision
0	H1: Income → Fashion Spending	0.0294	0.0142	Supported at 5%
1	H2: Purchase Frequency → Fashion Spending	15.1681	0.0960	Not supported at 5%
2	H3: Discount Influence → Fashion Spending	-13.9760	0.2185	Not supported at 5%
3	H4: Social Media Influence → Fashion Spending	12.3106	0.3087	Not supported at 5%
4	H5: Personalisation → Fashion Spending	-2.9044	0.8655	Not supported at 5%
5	H6: Brand Reputation → Fashion Spending	7.8039	0.6097	Not supported at 5%

45. Final mode

These are the headline statistics to use when writing the dissertation results section.

In [140]:
```
print("FINAL MODEL SUMMARY")
print("-------------------")
print(f"Observations: {int(model_m4.nobs)}")
print(f"R-squared: {model_m4.rsquared:.4f}")
print(f"Adjusted R-squared: {model_m4.rsquared_adj:.4f}")
print(f"F-statistic: {model_m4.fvalue:.4f}")
print(f"Model p-value: {model_m4.f_pvalue:.6f}")
print(f"AIC: {model_m4.aic:.2f}")
print(f"BIC: {model_m4.bic:.2f}")
print(f"Durbin-Watson: {dw_full:.3f}")
print(f"Shapiro-Wilk p-value: {shapiro_full.pvalue:.4f}")
```

```
FINAL MODEL SUMMARY
-------------------
Observations: 68
R-squared: 0.1543
Adjusted R-squared: 0.0711
F-statistic: 1.8552
Model p-value: 0.103278
AIC: 820.67
BIC: 836.21
Durbin-Watson: 2.476
Shapiro-Wilk p-value: 0.0008
```

In [141]:

```
significant = coef_table.drop(index='Intercept')
significant = significant[significant['p_value'] < 0.05]

print(
    f"The final model explains {model_m4.rsquared*100:.1f}% "
    f"of the variation in estimated monthly fashion spending."
)

if significant.empty:
    print("No individual predictor is statistically significant at
else:
    print("Statistically significant predictors at the 5% level:")
    for variable, row in significant.iterrows():
        direction = "positive" if row['Coefficient'] > 0 else "nega
        print(
            f"- {variable}: {direction} association; "
            f"coefficient={row['Coefficient']:.3f}, "
            f"p={row['p_value']:.4f}"
        )

print(
    "\nThese results should be interpreted as associations rather t
)
```

```
The final model explains 15.4% of the variation in estimated month
ly fashion spending.
Statistically significant predictors at the 5% level:
- Income_Numeric: positive association; coefficient=0.029, p=0.014
2

These results should be interpreted as associations rather than ca
usal effects.
```

Final conclusion

This notebook provides a complete quantitative analysis for the dissertation **Customer Behaviour & Personalisation in Fashion Retail**.

The analysis examined the factors influencing fashion spending among 68 consenting respondents. The study investigated income, purchase frequency, discount influence, social-media influence, personalisation and brand reputation to understand their relationship with estimated monthly fashion spending. The most important finding from the final multiple regression model is that income was the only statistically significant predictor of fashion spending. Income had a positive coefficient of 0.0294 ($p = 0.014$), indicating that respondents with higher income levels tended to report higher fashion spending. This relationship remained significant after controlling for purchase frequency, discount influence, social-media influence, personalisation and brand reputation. Therefore, income appears to be an important factor when identifying differences in customer spending behaviour. The other predictors were not statistically significant at the 5% level. Purchase frequency showed a positive relationship with spending ($\beta = 15.17$, $p = 0.096$), but the evidence was insufficient to classify it as statistically significant. Discount influence had a negative coefficient ($\beta = -13.98$, $p = 0.219$), while social-media influence had a positive coefficient ($\beta = 12.31$, $p = 0.309$). Neither relationship was statistically significant. Similarly, personalisation ($\beta = -2.90$, $p = 0.866$) and brand reputation ($\beta = 7.80$, $p = 0.610$) did not significantly predict fashion spending in the final model. The overall regression model produced an R^2 of 0.154, meaning that approximately 15.4% of the variation in fashion spending was explained by the six predictors. However, the adjusted R^2 was only 0.071, and the overall model was not statistically significant ($p = 0.103$). This indicates that the selected variables provide only limited explanatory power for overall fashion spending. Therefore, fashion retailers should not rely on these variables alone when predicting customer expenditure. Model comparison also showed that adding more variables did not consistently improve model performance. The full model had a higher R^2 than the earlier behavioural models, but its adjusted R^2 remained relatively low. This suggests that some of the additional variables may add complexity without providing sufficient additional explanatory power. The diagnostic analysis also identified an important limitation. The Shapiro-Wilk test was significant ($p = 0.0008$), indicating that the residuals do not fully satisfy the normality assumption. Therefore, the regression findings should be interpreted cautiously, particularly given the relatively small sample size. However, the VIF values for the explanatory variables were low, ranging from approximately 1.08 to 1.82, indicating that multicollinearity was not a major concern. Overall, the findings suggest that income is the strongest statistically supported factor associated with fashion spending in this sample. However, personalisation and brand reputation should not be concluded to have no business value simply because they were not significant predictors of spending. Their influence may operate through other outcomes such as purchase intention, customer experience or loyalty rather than directly determining spending. The results therefore support a more cautious and customer-focused approach to personalisation.